ECCV 2026 | MALMÖ | SEPT 8–12

3D Gaussian Splatting Compression with Object Scalability

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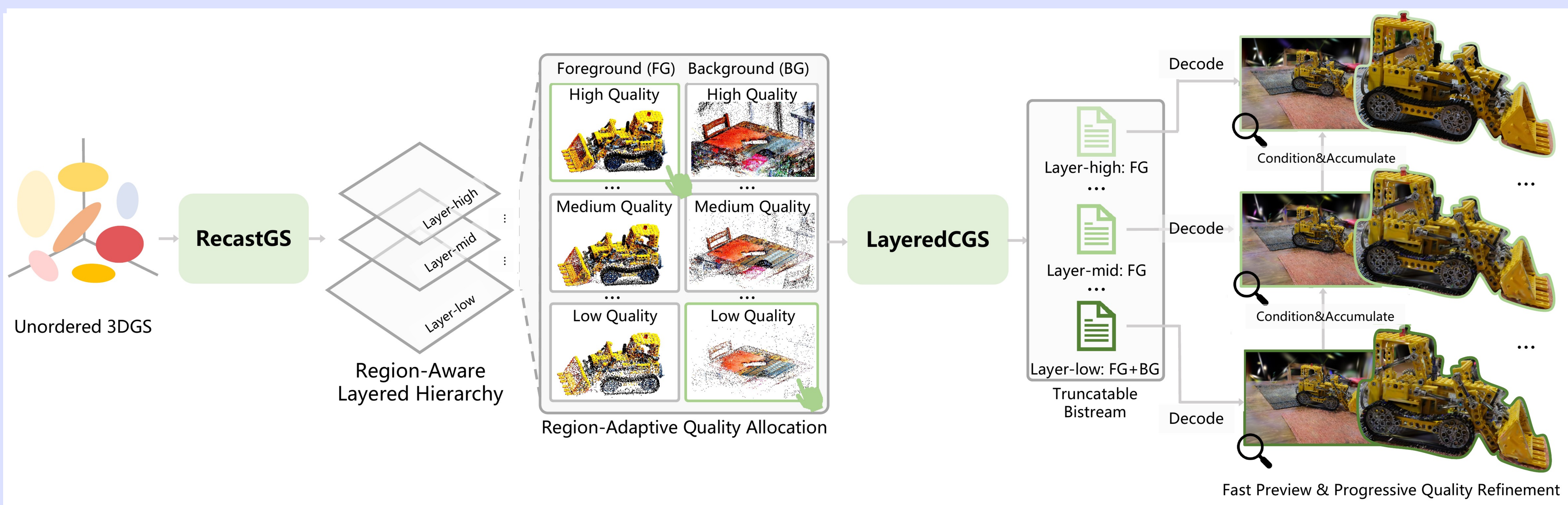
Task & Challenges



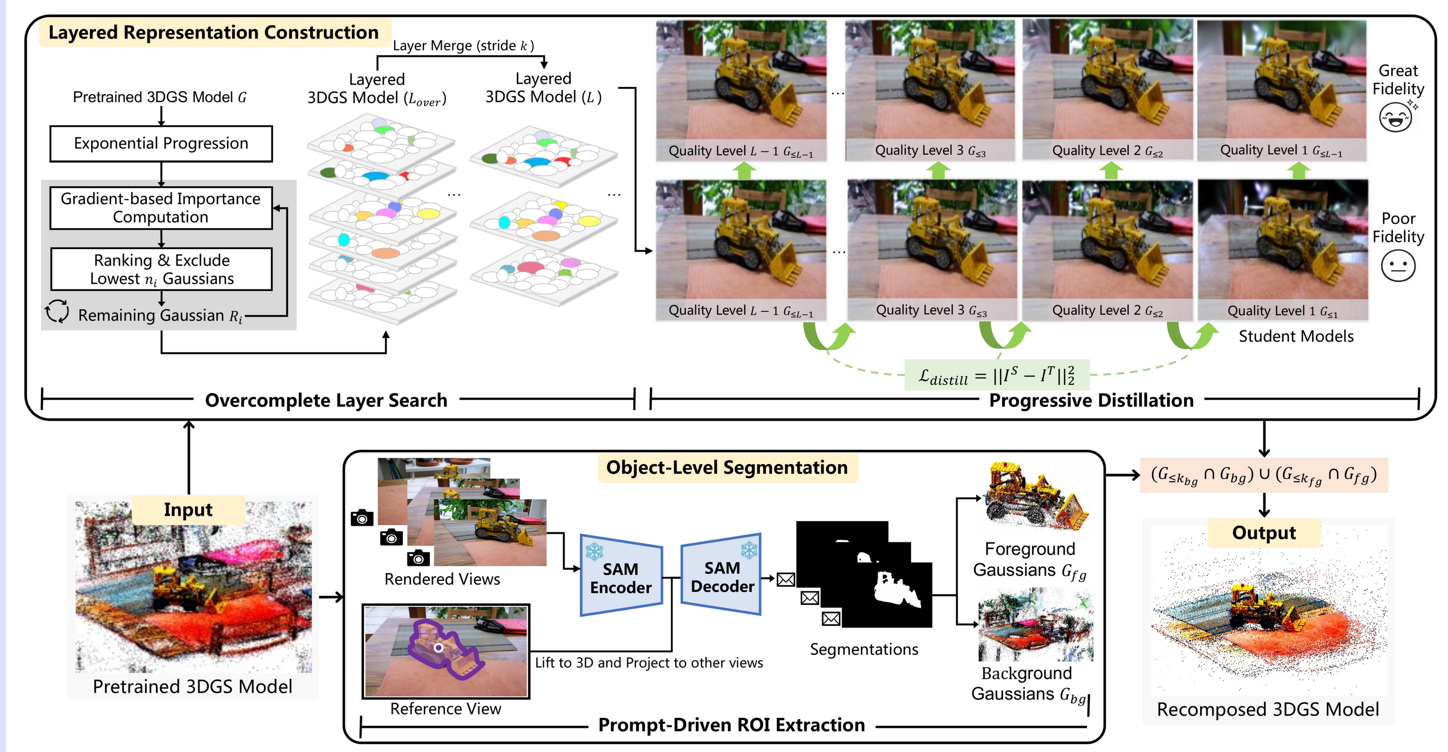
Task: Compress pretrained 3D Gaussian Splatting scenes while preserving interactive, object-level quality control.

Challenges:

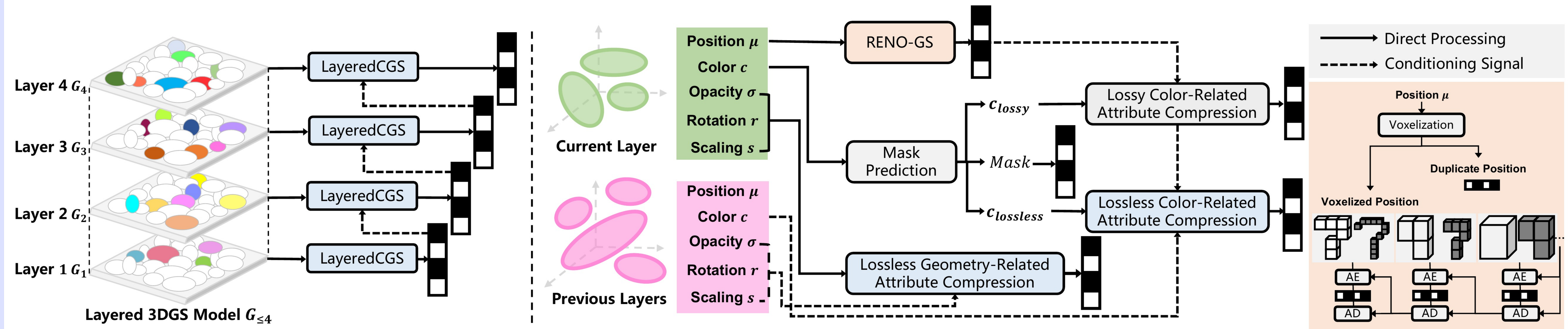
- Vanilla 3DGS has no intrinsic hierarchy, so global compression wastes bits on less important regions.
- Object-level regions of interest require recomposable representations rather than a single uniform-quality model.
- Feed-forward 3DGS codecs are fast, but holistic bitstreams are hard to truncate for preview and progressive refinement.

Core idea: **reorganize first, then compress layer by layer.**

Our Approach



RecastGS: reorganizes unordered 3DGS into a region-aware layered hierarchy via overcomplete search, progressive distillation, and prompt-driven object extraction.



LayeredCGS: compresses each layer conditioned on decoded lower layers, producing independently decodable bitstream segments for preview and progressive refinement.

Layered Representation

$$C_i = \left\lfloor n_1 \cdot \exp\left(\frac{\log N_g - \log n_1}{L-1}\cdot i\right)\right\rfloor, \quad i=1,\dots,L.$$

$$n_i = C_i - C_{i-1}, \quad i=2,\dots,L,$$

$$s_g = \sum_{v\in V}\sum_{p\in P(g)} \left\|\frac{\partial \mathcal{L}(v)}{\partial p}\right\|_2,$$

N_g: total Gaussians; L: number of layers
C_i: cumulative count up to layer i; n_i: count in layer i
s_g: importance score; V: training views; $P(g)$: parameters of g

Layer-wise Context

$$f_v = \frac{\sum_{k\in N(v)} w_k \hat{y}_k}{\sum_{k\in N(v)} w_k},$$

$$\mathcal{L} = \frac{bits_{total}}{N_g \times D},$$

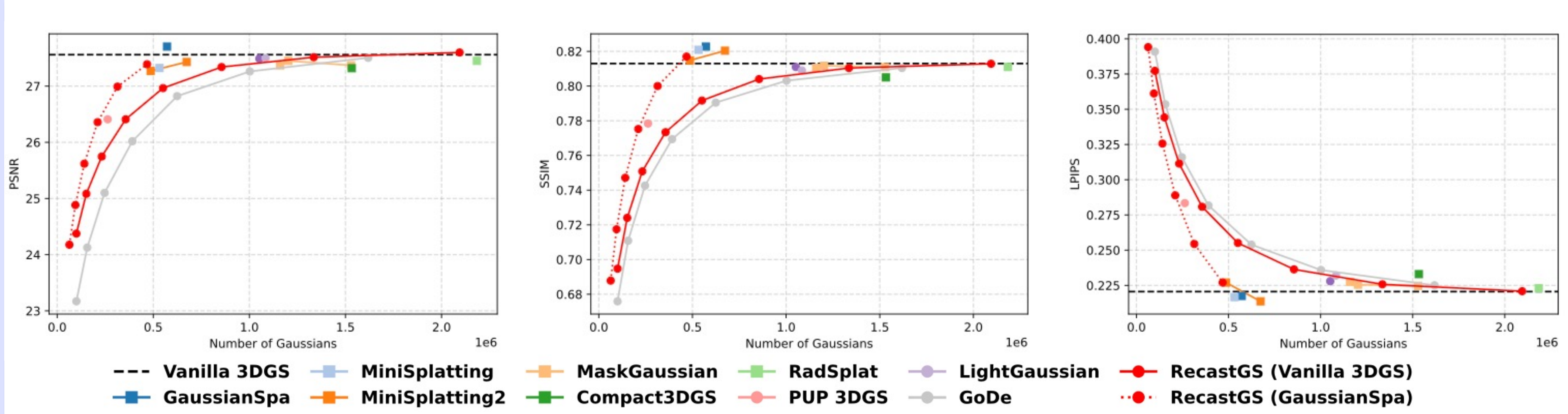
f_v: context feature at grid entry v
$N(v)$: decoded Gaussians contributing to v
w_k: distance-based interpolation weight;
$\hat{y}_k$: decoded latent feature
$\mathcal{L}$: rate loss; $bits_{total}$: total coded bits
N_g: number of Gaussians; D: attribute dimension

ROI Recomposition

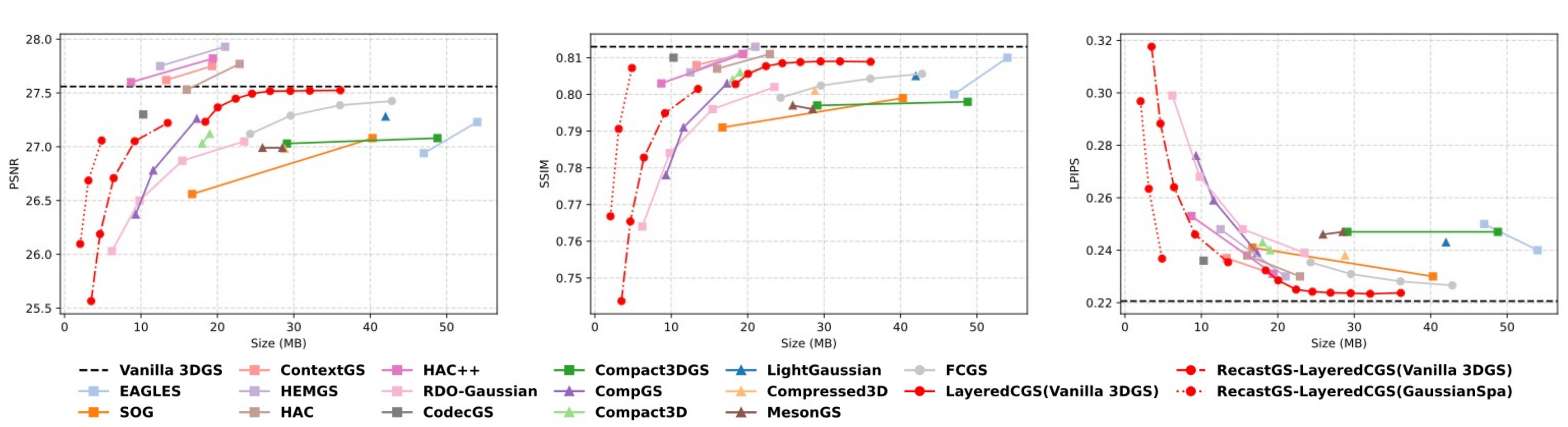
$$G^{rec} = (G_{\le k_{fg}} \cap G_{fg}) \cup (G_{\le k_{bg}} \cap G_{bg}).$$

G^{rec}: recomposed Gaussian set
G_{fg} / G_{bg} : foreground / background subsets
k_{fg} / k_{bg} : target quality levels for ROI control

Quantitative Comparison

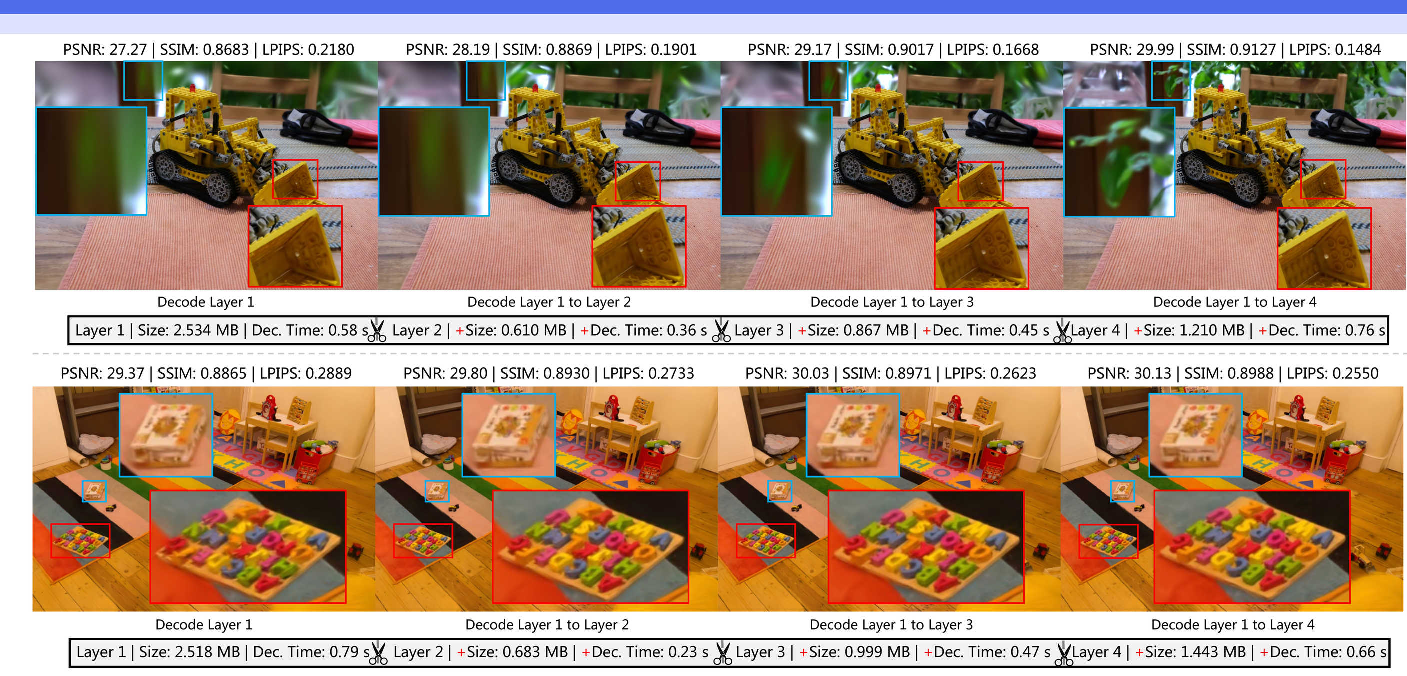


RecastGS reduces BD-BR vs. GoDe by 24.90% / 18.69% / 12.90% on PSNR / SSIM / LPIPS.



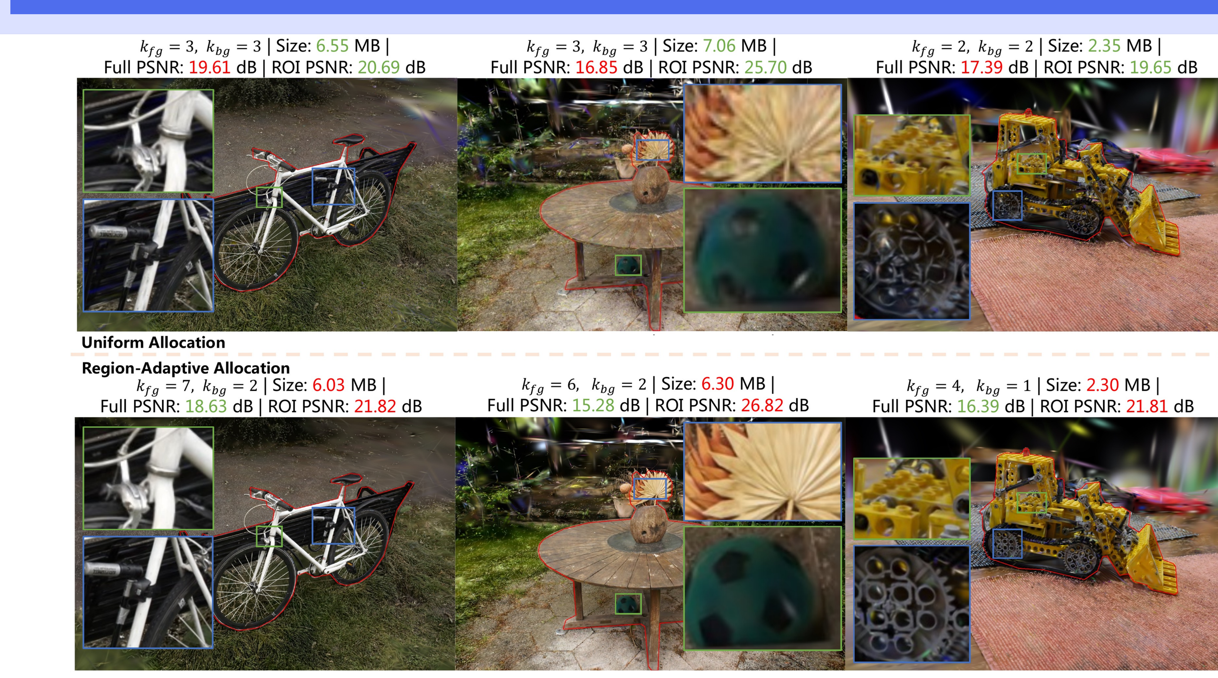
LayeredCGS reduces BD-BR vs. FCGS by 34.91% / 38.42% / 34.83% on PSNR / SSIM / LPIPS.

Progressive Decoding & Flexible Bitstream Truncation



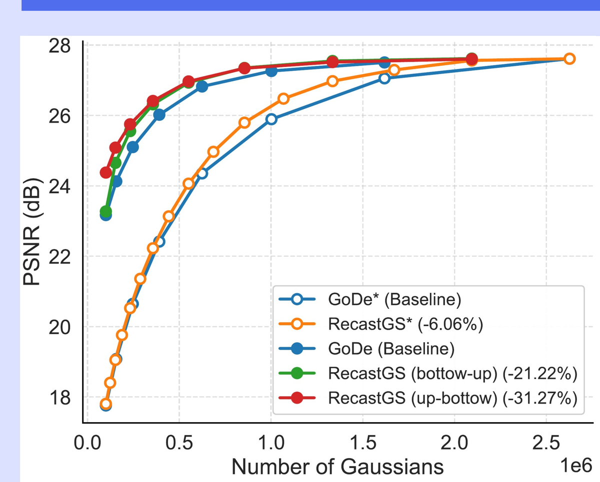
0.7 s coarse preview with progressive refinement.

ROI-aware Compression Performance

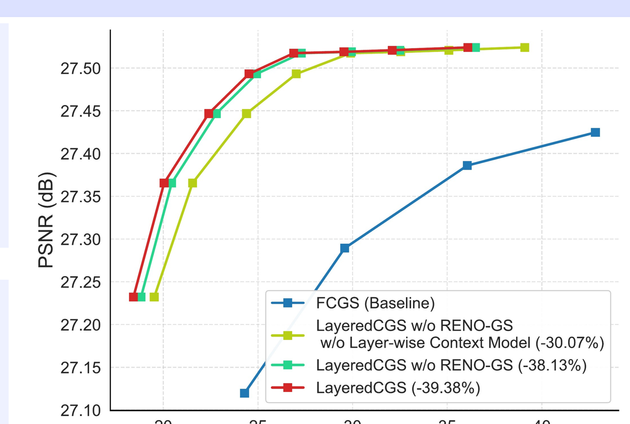


ROI allocation adds +2.16 dB ROI PSNR on kitchen.

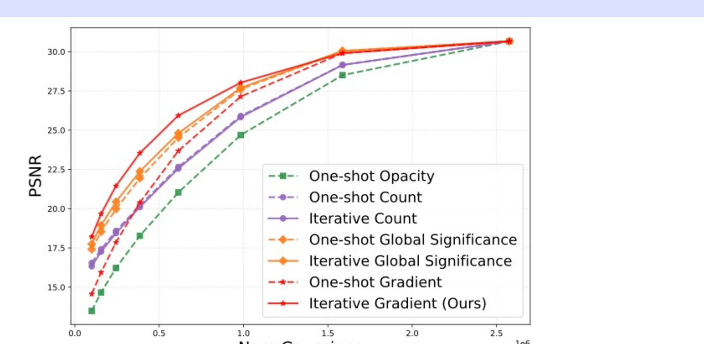
Ablation Study



Overcomplete Layer search: 6.06% Gain;
Progressive Distillation(bottow-up): 21.22% Gain;
Progressive Distillation(up-bottow): 31.27% Gain.



Layer-wise Context Model: 8.06% Gain;
RENO-GS: 1.25% Gain;
Decoding Speed:about 1.7x faster.



Comparison of different criteria and selection strategies used in Layered Representation.